

Vuk Aleksijević

Game Programmer, C++ and Unreal Engine

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Summary

Game programmer with 2+ years of production experience on Injustice 2 Mobile (credited), a live-service fighting game on iOS and Android. Single-handedly delivered 2 new playable characters, and implemented 10 artifacts and raid boss mechanics alongside store, game mode and online features, in Unreal Engine with C++ and Blueprints. Mentors a junior developer and takes part in technical hiring. Master's in Software Engineering (GPA 9.5/10), with a strong low-level foundation from building an OS kernel, a compiler and an assembler/linker toolchain.

Experience

Sperasoft, a Keywords Studio

Junior Software Developer II

Jan 2025 – present

Junior Software Developer

Aug 2024 – Jan 2025

Injustice 2 Mobile by NetherRealm Studios, shipped on iOS and Android. Worked in C++, Blueprints, UMG, TypeScript and Python.

Combat and gameplay

- Implemented 2 new playable characters independently, from the first gameplay pass to release: passives, special attacks, VFX setup, gear stat bonuses, post-playtest tuning and all follow-up fixes.
- Implemented 10 artifacts (equippable items with combat effects) and boss mechanics for 4 levels of a Solo Raid event, and playtested them with Game Design before release.
- Reworked abilities and legendary gear set bonuses for existing characters, and implemented the ability icon refactor on the combat HUD.
- Raised an infinite-loop risk in the forced tag-out rules with Game and Tech Design before implementation, so the edge case was settled in design instead of patched in code later.

UI, online and live-service features

- Built a new store offers system end to end: a loot table class extending the existing loot tables, offer menu, tiles, level preview and purchase flow in UMG/C++, online integration, per-vault purchase limits and a new vault key currency.
- Delivered the UI and online side of the Solo Raid rework: hub menu with a premium in-app purchase, training mode settings and pre-fight flow, info tutorial and a new achievement set.
- Restructured the main store with a side bar, new tabs and deep-link navigation, and built the sellback flow that converts duplicate exclusive rewards.
- Added player muting to in-game chat for customer support and bulk claim and repeat actions for Operations, and integrated UI with backend services for rewards, matchmaking, League Invasions and the Artifact Switcher.

Stability and tools

- Diagnosed and fixed a live crash affecting around 10,000 players.
- Wrote a Python tool that automates exporting loot tables to CSV, automated populating profile portraits in the sellback data, and wrote internal documentation used for onboarding.

Team

- Mentoring a junior developer: onboarded him to the Perforce workflow, the local backend environment, UMG layout practices and configuring server-side game states for testing.
- Interview candidates as part of the company's technical hiring process.
- Review code for correctness, performance and readability, and work daily with game designers, tech designers, UI artists and QA, including walking QA through the combat damage calculations.

Internal game development bootcamp at Sperasoft

Tower defense game in Unreal Engine and C++.

- Built enemy AI with behavior logic, a modular quest system and a save-game system in C++.
- Implemented core gameplay systems (pause, timers, slow motion) and contributed to level and gameplay design.

Projects

Top-down action game

Unreal Engine 4 and C++. In development, co-developed with a friend.

- Game built around a hideout hub with interactive stations and a raid loop. Gameplay code in C++ with Blueprint subclasses, using Enhanced Input, Gameplay Tags and custom subsystems for raid timing, time of day and widget management.

Game jams

Godot and GDScript. Four-person teams, main programmer.

- **Not the Mosquito Again**, multiplayer game Ludum Dare 56
- **HoverBald**, procedurally generated obstacle-avoidance game Retro Game Jam 2024
- **Fauxtelja**, strategy and resource-management game SKC Game Jam 2025

Android apps

Kotlin, Jetpack Compose, Room, MVVM.

- **Adaptive-speed RSVP reading app** (master's thesis). Displays text word by word with per-word timing adapted to word length and punctuation, imports PDFs and web articles, and includes speed calibration and a self-paced reading test with PDF reports. Evaluated in a user study with 9 participants against reading on paper.
- **TenThousand**. Habit and focus-timer app with a gacha system for unlocking animated, shader-based (AGSL) backgrounds.

Academic systems projects

- **Operating system kernel for RISC-V** (C++, RISC-V assembly): threads, memory management and time-sharing.
- **MicroJava compiler** (Java): complete compiler from lexical analysis to code generation.
- **Assembler, linker and emulator** for a custom architecture.
- **CPU design** in VHDL/Verilog, verified on FPGA.

Skills

Languages	C++, C, Python, TypeScript, Kotlin, Java, GDScript, SQL
Engines	Unreal Engine 4/5 (C++, Blueprints, UMG, Enhanced Input, Gameplay Tags, Subsystems), Godot
Game systems	Combat mechanics and status effects, boss mechanics, gameplay and HUD UI, store and economy systems, online integration, enemy AI, save systems
Tools	Perforce, Git, Jira, Visual Studio, CLion, Android Studio
Other	Jetpack Compose, Room, TensorFlow/Keras, Flask, Node.js, REST APIs

Education

University of Belgrade, School of Electrical Engineering

Master's Degree, Software Engineering GPA 9.5/10 2026
Thesis: *Reading support application using adaptive speed and RSVP text display* (Android).

Bachelor's Degree, Computer Engineering and Information Theory 2024
Thesis: *Food image classification using transfer learning*. Fine-tuned an ImageNet-pretrained InceptionV3 model (TensorFlow/Keras) on a 101-class food dataset and served predictions through a Flask REST API with a web front end.

Additional

Training	Effective Communication, Leadership (internal Sperasoft courses)
Languages	Serbian (native), English (fluent, bilingual English-track high school), Italian (professional working proficiency)
Interests	Video games, drawing, cycling, gym, cooking